

Atoms in DFT Report

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1 Introduction

The first part of the lab work on computational physics is focus on the Calculation of Electronic Energy levels of differnt atoms such as Helium, carbon, Nitrogen and oxygen with VASP. This type of software takes the calculations inputs from a few standard files :

- POSCAR : contains the information about the lattice and the position of the atoms in the unit cell.
- POTCAR : contains the information about the pseudopotential of the atoms in the system.
- INCAR : contains the information about the settings of the calculation such as the type
- KPOINTS : contains the information about the k-points grid used for the calculation.
- OUTCAR : contains the output of the calculation such as the energy levels and the forces on the atoms.

Fot this calculation these are the only required even if there are other files that can be used for more specific calculations.

2 Calculations Method

After affect test about He atom with a fixed lattice parameter,
we performed the calculations for the other atoms with the same settings.

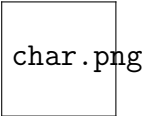


Figure 1: Resonance Characteristic Curve of the NCHR cantilever. Notice the unusual difference in the phase plot, an error must have occured during data acquisition. Amplitude modulus set to 20nm, set point 15.1nm.

3 Observations and Results

Key Parameters	Image Size	Scan Rate	Set Point	Drive Amplitude
	15x15 μm	0.5 Hz	15.1 nm	20 nm

Table 1: Main parameters used during NCM-AFM scan are shown here. Some info could be missing due to data error explained before.

4 Final Results

While the experiment seemed to be performed without any important issues during the AFM operation and data acquisition in the laboratory, the afterwards analysis of the collected data was quite problematic. On a good note the available results were as predicted, however the details were insufficient to pursue a complete evaluation.

Results for the resonance curve were consistent for the amplitude but a significant shift was noticed on

the phase curve. This can be due to different reasons, with the most probable one being bad settings of the software during acquisition.

The results of the NCM-AFM imaging were fine but no height scale bar was acquired. Finally, for the AFM Spectroscopy the results were as predicted from the theoretical curves with the Attractive regime approximation.